

Chapter 14

§14.1.

function of two or more variables.

$$f(x, y), f(x, y, z).$$

generally, $f(\underbrace{x_1, x_2, \dots, x_n}_{n\text{-tuple of real numbers}})$

domain: $D \subseteq \mathbb{R}^2$

range: $\{f(x, y) \mid (x, y) \in D\}$. } for $z = f(x, y)$.

Level curves

$f(x, y) = k$ where k is a constant.

§14.2.

Limit of a function of two variables

$$\lim_{(x, y) \rightarrow (a, b)} f(x, y) = L$$

Remark: If $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along a path C_1 and $f(x, y) \rightarrow L_2$ as $(x, y) \rightarrow (a, b)$ along another path C_2 , where $L_1 \neq L_2$, then $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$ does not exist.

* f is continuous at (a, b) iff $\lim_{(x, y) \rightarrow (a, b)} f(x, y) = f(a, b)$.

§14.3.

First partial derivatives

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \leftarrow \text{regard } y \text{ as a constant.}$$

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} \leftarrow \text{regard } x \text{ as a constant.}$$

$$\text{notations: } f_x(x, y) = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = D_x f$$

$$f_y(x, y) = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = D_y f$$

Higher partial derivatives

$$\begin{aligned} (f_x)_x &= f_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) & (f_x)_y &= f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) \\ (f_y)_x &= f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) & (f_y)_y &= f_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) \end{aligned} \quad \left. \vphantom{\frac{\partial^2 f}{\partial y \partial x}} \right\} \text{second derivatives}$$

In general

$$u = f(x_1, x_2, \dots, x_n)$$

$$\rightarrow \frac{\partial f}{\partial x_i} = f_{x_i} = D_i f = \frac{\partial u}{\partial x_i}, \quad i=1, 2, \dots, n.$$

Clairaut's Theorem

If f_{xy} and f_{yx} are both continuous at (a, b) , then $f_{xy}(a, b) = f_{yx}(a, b)$.

§ 14.5

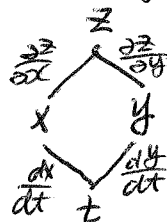
Chain Rules

Case I:

If $z = f(x, y)$ with $x = g(t)$ and $y = h(t)$, then

$$\boxed{\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = f_x g' + f_y h'}$$

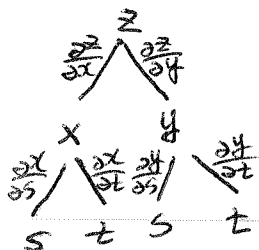
Tree diagram



Case II:

If $z = f(x, y)$ with $x = g(s, t)$ and $y = h(s, t)$, then

$$\boxed{\begin{aligned} \frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} & \frac{\partial z}{\partial t} &= \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t} \\ &= f_x g_s + f_y h_s & &= f_x g_t + f_y h_t \end{aligned}}$$



General form

$$u(x_1, x_2, \dots, x_n), \quad x_j(t_1, t_2, \dots, t_m) \text{ for } j=1, 2, \dots, n$$

then

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_n} \frac{\partial x_n}{\partial t_i} \quad i=1, 2, \dots, m.$$

Implicit differentiation

$F(x, y) = 0 \rightarrow y = f(x)$ is implicitly defined

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} = 0$$

$$\frac{\partial y}{\partial x} = - \frac{\partial F / \partial x}{\partial F / \partial y} = - \frac{F_x}{F_y}$$

§14.4.

Given a surface $S (z = f(x, y))$, its tangent plane at the point $(x_0, y_0, \overset{f(x_0, y_0)}{z_0})$ is defined by

$$z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$$

Linear Approximation

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) =: L(x, y).$$

differentials $z = f(x, y)$

$$dz = f_x(x, y)dx + f_y(x, y)dy.$$

total differentials. $\rightarrow f(x+dx, y+dy) \approx f(x, y) + dz$

§14.6.

The directional derivative of f at (x, y) in the direction of a unit vector $u = \langle u_1, u_2 \rangle$ is

$$D_u f(x, y) = \lim_{h \rightarrow 0} \frac{f(x+hu_1, y+hu_2) - f(x, y)}{h}$$

$$= \langle f_x(x, y), f_y(x, y) \rangle \cdot \langle u_1, u_2 \rangle$$

$\nabla f(x, y)$ — Gradient vector

$$= \underline{\nabla f \cdot u} = f_x u_1 + f_y u_2$$

Function of three variables: $D_u f(x, y, z) = \langle f_x, f_y, f_z \rangle \cdot \langle u_1, u_2, u_3 \rangle$
 $= \nabla f \cdot u = f_x u_1 + f_y u_2 + f_z u_3$

$\max_u D_u f = |\nabla f| \rightarrow$ when take u be the same direction as ∇f .

Tangent planes to level surfaces

Level surface: $F(x, y, z) = k$

At (x_0, y_0, z_0) , normal vector is given by $\nabla f(x_0, y_0, z_0)$, thus the equation of the tangent plane to the level surface is given by

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

Normal line - symmetric equations

$$\frac{x - x_0}{F_x(x_0, y_0, z_0)} = \frac{y - y_0}{F_y(x_0, y_0, z_0)} = \frac{z - z_0}{F_z(x_0, y_0, z_0)}$$

§14.7.

local maximum/minimum, absolute maximum/minimum.

Critical points (or stationary points): $f_x(a, b) = 0, f_y(a, b) = 0$
 $\Rightarrow (a, b)$ is a critical point.

First derivative Test:

If f has a local maximum/minimum at (a, b) , then $f_x(a, b) = f_y(a, b) = 0$,
i.e., (a, b) is a critical point.

Second derivative Test

Suppose $f_x(a, b) = f_y(a, b) = 0$. Let $D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$.

(a) If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.

(b) If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.

(c) If $D < 0$, then $f(a, b)$ is neither a local maximum nor minimum,
 (a, b) is called a saddle point.

§14.8.

$$\begin{cases} \max/\min f(x, y, z) \\ \text{s.t. } g(x, y, z) = k \\ \quad \downarrow \text{constraint.} \end{cases}$$

Method of Lagrange multipliers

(a) Find all values of x, y, z and λ such that $\nearrow$ multiplier

$$\begin{cases} \nabla f(x, y, z) = \lambda \nabla g(x, y, z) \\ g(x, y, z) = k \end{cases} \quad \text{i.e.} \quad \begin{cases} f_x = \lambda g_x \\ f_y = \lambda g_y \\ f_z = \lambda g_z \\ g = k \end{cases}$$

(b). Evaluate f at all points resulted from (a). The largest is the maximum value of f and the smallest is the minimum value of f .

For two constraints problem

$$\begin{cases} \max/\min f(x, y, z) \\ \text{s.t. } g(x, y, z) = k_1 \\ \quad h(x, y, z) = k_2 \end{cases}$$

$$\Rightarrow \begin{cases} \nabla f = \lambda \nabla g + \mu \nabla h \\ g = k_1 \\ h = k_2 \end{cases}$$

λ, μ two multipliers.

Chapter 15.

§15.1.

The double integral of f over the rectangle R is

$$\iint_R f(x,y) dA = \lim_{m,n \rightarrow \infty} \underbrace{\sum_{i=1}^m \sum_{j=1}^n f(x_i^*, y_j^*) \Delta A}_{\substack{\text{double Riemann sum} \\ \Delta A = \Delta x \Delta y}}$$

If $f(x,y) \geq 0$, the volume V of the solid that lies above R and below the surface $z = f(x,y)$ is

$$V = \iint_R f(x,y) dA.$$

Midpoint Rule:

$$\iint_R f(x,y) dA \approx \sum_{i=1}^m \sum_{j=1}^n f(\bar{x}_i, \bar{y}_j) \Delta A.$$

Properties

$$\iint_R [f + g] dA = \iint_R f dA + \iint_R g dA$$

$$\iint_R \alpha f dA = \alpha \iint_R f dA \quad \text{where } \alpha \text{ is a constant.}$$

§15.2.

Fubini's Theorem

$R = \{(x,y) \mid a \leq x \leq b, c \leq y \leq d\}$, f is continuous over R , then

$$\iint_R f(x,y) dA = \int_a^b \int_c^d f(x,y) dy dx = \int_c^d \int_a^b f(x,y) dx dy.$$

called "iterated integral"

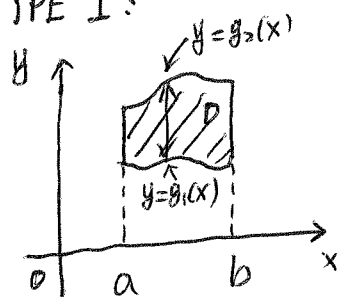
special case $f(x,y) = g(x)h(y)$ on $R = [a,b] \times [c,d]$

$$\iint_R f(x,y) dA = \int_a^b g(x) dx \int_c^d h(y) dy.$$

§15.3.

TYPE of Planar regions

TYPE I:

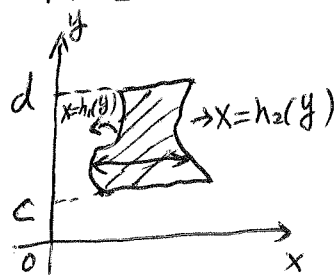


$$D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

$$\Downarrow$$

$$\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$$

TYPE II:



$$D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$$

$$\Downarrow$$

$$\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$$

Remark: The plane region D could be both Type I and Type II sometimes.
Always use the one that gives simpler integration formula.

Properties

(i) If $D = D_1 \cup D_2$ and $D_1 \cap D_2 = \emptyset$, then

$$\iint_D f dA = \iint_{D_1} f dA + \iint_{D_2} f dA$$

(ii) If $f \geq g$ on D , then

$$\iint_D f dA \geq \iint_D g dA$$

(iii) $\iint_D 1 dA = \text{Area}(D)$, where D is a plane region.

(iv) If $m \leq f(x, y) \leq M$ on D , then

$$m \text{Area}(D) \leq \iint_D f(x, y) dA \leq M \text{Area}(D)$$

$\rightarrow$ used for a rough estimation of the double integral $\iint_D f(x, y) dA$.